LAB RECORD

BCSC0806 MICROPROCESSOR LAB

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UNIV. ROLL NO.: 2015004S1

BRANCH: CS

BATCH: C2

SESSION: 2021 - 22

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BCSC0806: MICROPROCESSOR LAB

Semester: IV

L–T–P-J: 0–0–2-0

Credits: 01

INDEX

S. No.	Experiment	Date of Perform	Date of Submission	Grading scale(10)	Signature
1	Write an assembly language program to store 8 bit data into memory.	16th Feb'22		7-	
2	write an assembly language program to add exchange the content of 2 memory locations	16th Feb'22		7	
3	Write an assembly language program to add two 8 bit numbers	16th Feb'22		7+	13/4
4	write an assembly language program to ~~sub~~ two 8-bit numbers	16th Feb'22		B-	
5	write an assembly language program to add two 16 bit numbers	16th Feb'22		B-	
6	Write an assembly language program to sub two 16 bit numbers	16th Feb'22		B	13/4
7	Write an assembly language program to find 1's complement of 8 bit number.	23rd Feb'22		B	
8	Write an assembly language program to find 2's complement of 8 bit number.	23rd Feb'22		B+	
9	Write an assembly language program to find 1's complement of 16 bit number	23rd Feb'22		B+	
10	Write an assembly language program to find 2's complement of 16 bit number.	23rd Feb'22		B++	13/4
11	Write an assembly language program to for pack 2 8 bit numbers.	23rd Feb'22		B++	
12	Write an assembly language program to unpack 2 8 bit numbers	23rd Feb'22		A	
13	Write an assembly language program to find whether a no is -ve or positive.	9th Mar'22		A	13/4
14	Write an assembly language program to find -ve no in a block of 10 numbers	9th Mar 9 Feb'22		A	

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Semester: IV **L–T–P-J: 0–0–2-0** **Credits: 01**

INDEX

S. No.	Experiment	Date of Perform	Date of Submission	Grading scale(10)	Signature
15	Write an assembly language program to find the sum of series of 10 numbers.	23rd Mar' 2022		2	
16	Write an assembly language program to find no of ones in a byte	23rd Mar 2022		2	Dw 13/4
17	Write an assembly language program to find largest no in an array.	30th Mar 2022		2	
18	Write an assembly language program to add 2 8 bit numbers considering carray.	30th Mar 2022		2	
19	Write an assembly language program to multiply 2 8 bit numbers.	6th April 2022		2	Dw 13/4
20	Write an assembly language program to count even numbers from a series.	6th April 2022		2	
21	WAP to arrange an array in ascending order	6th April 2022		2	
22	WAP to left shift a 8 bit number.	13th April 2022		2	Dw 21/4
23	WAP to check whether a number is even or odd.	13th April 2022		2	
24	WAP to check whether a number is positive or negative	13th April 2022		2	
25	Write an assembly language program to count odd numbers from a series	13th April 2022		2	Dw 21/4
26	write an assembly language program to find smallest numbers from an array			2	

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Experiment No. 1

Objective: Write an assembly language program to store 8 bit data into memory.

Flow Chart (or) Algorithm:

1) Move data into accumulator.
2) Store the data of accumulator at a memory location.
3) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	MVI A	3E	
2	2001		0B	
3	2002	STA	32	
4	2003		00	
5	2004		25	
6	2005	HLT	76	

Input/Output: Input : 08

Output : '08' stored at 2500

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Experiment No. 2

Objective: To exchange the contents of two memory locations.

Flow Chart (or) algorithm:

1) Load the accumulator with the content of one location.
2) Move the content of accumulator to another register.
3) Load the accumulator with the content of second location
4) Store at the content of accumulator at first memory location.
5) Move the content of previous register to accumulator.
6) Store the content of accumulator at second memory location.
7) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	LDA	3A	
2	2001		00	
3	2002		25	
4	2003	MOV B, A	47	
5	2004	LDA	3A	
6	2005		01	
7	2006		25	
8	2007	STA	32	
9	2008		00	
10	2009		25	
11	200A	MOV A, B	78	
12	200B	STA	32	
13	200C		01	
14	200D		25	
15	200E	HLT	76	
	200F			

Input/Output: Input : 2500 - 15 ; 2501 - 17

Output : 2500 - 17 ; 2501 - 15

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Experiment No. 3

Objective: Write an assembly language program to add two 8 bit numbers

Flow Chart (or) algorithm:

1) Move the data 1 into accumulator.
2) Move data 2 in a general purpose register.
3) Add the content of accumulator to the content of general purpose register.
4) Store the result into memory.
5) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	MVI A	3E	
2	2001		05	
3	2002	MVI B	06	
4	2003		04	
5	2004	ADD B	80	
6	2005	STA	32	
7	2006		50	
8	2007		20	
9	2008	HLT	~~76~~	

Input/Output: Input : 05 , 04

Output : 09 stored at location 2050

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Experiment No. 4

Objective: Write an assembly language program to subtract two eight bit numbers

Flow Chart (or) algorithm:

1) Move a number into accumulator.
2) Move second number into a general purpose register.
3) Subtract the content of accumulator and register.
4) Store the result in memory.
5) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	MVI A	3E	
2	2001		05	
3	2002	MVI B	06	
4	2003		04	
5	2004	SUB B	90	
6	2005	STA	32	
7	2006		50	
8	2007		80 20	
9	2008	HLT	76	

Input/Output: Input : 05, 04

Output : 01 stored at 2050

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Experiment No. 5

Objective: Write an assembly language program to add two 16 bit numbers.

Flow Chart (or) algorithm:

1) Load the content from memory (16 bit number) into HL pair register.
2) Load the content from memory (16 bit number) into DE pair register.
3) Move the content of L to ~~ad~~accumulator.
4) Add the content of E to accumulator.
5) Store the result in memory.
6) Move the content of register H into accumulator.
7) Add the content of accumulator to register D.
8) Store the result into memory.
9) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	LXIH	21	
2	2001		54	
3	2002		32	
4	2003	LXID	11	
5	2004		94	
6	2005		58	
7	2006	MOV A,L	7D	
8	2007	ADD E	83	
9	2008	STA	32	
10	2009		00	
11	200A		25	
12	200B	MOV A,H	7C	
13	200C	ADC D	8A	
14	200D	STA	32	
15	200E		01	
16	200F		25	
17	2010	HLT	76	

Input/Output: Input : HL : 3254
DE : 5894
Output: HL : 8AE8

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Experiment No. 6

Objective: Write an assembly language program to subtract two 16 bit numbers

Flow Chart (or) algorithm:

1) Move a 16 bit number from memory to HL pair register.
2) Move second 16 bit number from memory to DE pair register.
3) Move the content of register L to accumulator.
4) Subtract the contents of register E and accumulator.
5) Store the result in memory.
6) Move the content of register H to accumulator.
7) Subtract the contents of register D and accumulator.
8) Store the result in memory.
9) Stop.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	LXIH	21	
2	2001		54	
3	2002	~~LXID~~	32	
4	2003	LXID	11	
5	2004		94	
6	2005		58	
7	2006	MOV A,L	7D	
8	2007	SUB E	93	
9	2008	STA	32	
10	2009		00	
11	200A		25	
12	200B	MOV A,H	7C	
13	200C	SBB D	9A	
14	200B	STA	32	
15	200E		01	
16	200F		25	
17	2010	HLT	76	

Input/Output: Input: 3254 (HL), 5894 (DE)

Output: D9C0 (HL)

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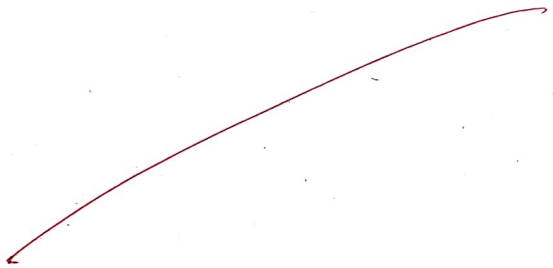
Experiment No. 7

Objective: Write an assembly language program to find 1's compliment of 8 bit number.

Flow Chart (or) algorithm:

1) Move a 8 bit number into accumulator.
2) Complement the number.
3) Store in memory
4) Stop



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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	MVI A	3E	
2	2001		05	
3	2002	CMA	2F	
4	2003	STA	32	
5	2004		00	
6	2005		25	
7	2006	HLT	76	

Input/Output: Input : 05

Output : FA stored at memory location 2500.

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Experiment No. 8

Objective: Write an assembly language program to find 2's compliment of 8 bit number.

Flow Chart (or) algorithm:

1) Move a number into accumulator directly.
2) Complement the number.
3) Add 1 to the complement.
4) Store the result into memory
5) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	MVI A	3E	
2	2001		05	
3	2002	CMA	2F	
4	2003	ADI	C6	
5	2004		01	
6	2005	STA	32	
7	2006		00	
8	2007		25	
9	2008	HLT	76	

Input/Output: Input : 05
Output : FB

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Experiment No. 9

Objective: Write an assembly language program to find 1's complement of 16 bit number

Flow Chart (or) algorithm:

1) Load the data into HL pair register.
2) Move the data of register L to accumulator.
3) Complement the content of accumulator.
4) Move the data of accumulator into register L.
5) Move the data of register H to accumulator.
6) Complement the content of accumulator.
7) Move the content of accumulator into register H.
8) Store the contents of HL pair register into memory.
9) Stop.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	LXIH	21	
2	2001		45	
3	2002		13	
4	2003	MOV A,L	~~137D~~	
5	2004	CMA	2F	
6	2005	MOV L,A	6F	
7	2006	MOV A,H	7C	
8	2007	CMA	2F	
9	2008	MOV H,A	67	
10	2009	SHLD	22	
11	200A		00	
12	200B		25	
13	200C	HLT	76	
~~14~~	~~200D~~			
~~15~~	~~200E~~			
~~16~~	~~200F~~			
~~17~~	~~2010~~			

Input/Output: Input : 1345 (HL)
Output ECBA (2500)

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Experiment No. 10

Objective: Write an assembly language program to find 2's complement of 16 bit number.

Flow Chart (or) algorithm:

1) Load the 16 bit number in HL pair register.
2) Move the lower nibble from L to accumulator.
3) Complement the content of accumulator.
4) Move the content of accumulator to register L.
5) Move the content of register H to accumulator.
6) Complement the content of accumulator.
7) ~~Increase~~ Move the content of accumulator to register H.
8) Increment the content of H by 1.
9) Store the number in HL pair register into memory.
10) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	LXIH	21	
2	2001		45	
3	2002		13	
4	2003	MOV A,L	7D	
5	2004	CMA	2F	
6	2005	MOV L,A	6F	
7	2006	MOV A,H	7C	
8	2007	CMA	2F	
9	2008	MOV H,A	67	
10	2009	INXH	23	
11	200A	SHLD	22	
12	200B		00	
13	200C		25	
14	200D	HLT	76	

Input/Output: Input : 1345
Output : ECBB

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Experiment No. 11

Objective: Write an assembly language program to pack two BCD 8 bit numbers.

Flow Chart (or) algorithm:

1) Move the data into accumulator.
2) Move second data into register B.
3) Rotate the content of accumulator 4 times towards left.
4) Add the content of B to the content of accumulator.
5) Store the result in memory.
6) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	MVI A	3E	
2	2001		05	
3	2002	MVI B	06	
4	2003		04	
5	2004	RLC	07	
6	2005	RLC	07	
7	2006	RLC	07	
8	2007	RLC	07	
9	2008	ADD B	88	
10	2009	STA	32	
11	200A		00	
12	200B		25	
13	200C	HLT	76	

Input/Output: Input :- 05
04
Output :- 54

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Experiment No. 12

Objective: Write an assembly language program to unpack ~~two~~ 8 bit numbers.

Flow Chart (or) algorithm:

1) Move the BCD number to accumulator.
2) Copy the number to register B.
3) AND the content of accumulator with number F0.
4) Rotate the number towards right 4 times.
5) Store the result in memory.
6) Move the number from B to accumulator.
7) And the content of accumulator with number F0 0F.
8) Store the result in memory.
9) Stop.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	2000	MVI A	3E	
2	2001		54	
3	2002	MOV B, A	47	
4	2003	ANI	E6	
5	2004		F0	
6	2005	RRC	0F	
7	2006	RRC	0F	
8	2007	RRC	0F	
9	2008	RRC	0F	
10	2009	STA	32	
11	200A		00	
12	200B		25	
13	200C	MOV A, B	78	
14	200D	ANI	E6	
15	200E		0F	
16	200F	STA	32	
17	2010		01	
18	2011		25	
19	2012	HLT	76	

Input/Output:

Input :- 54

Output :- 05, 04

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Experiment No. 13

Objective: Write an assembly language program whether a number is negative or positive. Store the negative number or positive number.

Flow Chart (or) algorithm:

1) Move the number into accumulator
2) AND the number with number 80.
3) If the answer is zero, ie the number is positive, jump to label and store the number.
4) If the answer is not zero, ie the number is negative, store the number without jumping to label.
5) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
1	3000	MVIA	3E	
2	3001		FB	
3	3002	ANI	E6	
4	3003		80	
5	3004	JZ	CA	Label (300A)
6	3005		0A	
7	3006		30	
8	3007	STA	32	
9	3008		00	
10	3009		25	
11	300A	STA	32	(label location)
12	300B		01	
13	300C		25	
14	300D	HLT	76	

Input/Output: Input : FB
Output : No is negative therefore it will be stored at location 2501.

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Experiment No. 14

Objective: Write an assembly language program to find negative number in a block of 10 numbers.

Flow Chart (or) algorithm:

1) Move the number '10' for counter in general purpose register C.
2) Move another counter number in general purpose register B.
3) Load memory address of starting index of array in HL pair register.
4) Move the first number in accumulator.
5) And the number with '80'.
6) If the result is zero, jump to label.
7) If the result is not zero, increase the counter in B.
8) Else, increase the address of memory (increase the content of register H) if the result is zero.
9) Next, decrease the counter in C.
10) If counter in C is not zero, repeat the process
11) ~~If the number is negative, it move it into accumulator~~ Move value of B in accumulator
12) Store the content of accumulator at memory location. (No of negative numbers present in array are stored at memory address specified).
13) Stop.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	3000	MVIC ,0A	0E	
	3001		0A	
	3002	MVIB ,00	06	
	3003		00	
	3004	LXIH 2000	21	
	3005		00	
	3006		20	
back	3007	MOV A,M	7E	
	3008	ANI ,80	E6	
	3009		80	
loop	300A	JZ (label) 300E	CA	
	300B		0E	
	300C		30	
	300D	INX B	03	
label	300E	INX H	23	
	300F	DCR C	0D	
	3010	JNZ (back) 3007	C2	
	3011		07	
	3012		30	
	3013	MOV A,B	78	
	3014	STA 2500	32	
	3015		00	
	3016		25	
	3017			
	3018	HLT	76	

Input/Output:

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WAP to find sum of series of 10 no.

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Experiment No. 15

Objective: Write an assembly language program to find the sum of series of 10 numbers.

Flow Chart (or) algorithm:

1) Take a counter in a general purpose register, say C.
2) Take the starting address of series in HL pair register.
3) Move the content of HL pair register to accumulator.
4) Increement the H register's content
5) Add it to accumulator.
6) Decrease the counter value by 1.
7) If the answer is non-zero, jump to step 4.
8) Repeat until zero is encountered.
8) Store the sum into memory.
9) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	MVIC, 09	0E	
	2001		09	
	2002	LXIH 3000	21	
	2003		00	
	2004		30	
	2005	MOV A, M	7E	
	2006	INXH	24	
	2007	ADD M	86	
	2008	DCR C	0D	
	2009	JNZ 2006	C2	
	200A		06	
	200B		20	
	200C	STA 2500	32	
	200D		00	
	200E		25	
	200F	HLT	76	

Input/Output:

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Experiment No. 16

Objective: Write an assembly language program to find no of ones in a byte.

Flow Chart (or) algorithm:

1) Load data in accumulator from memory.
2) Move / initialize register B with data '08'.
3) Initialize register C with data '08'.
4) Rotate contents of A with carry to left.
5) Jump if no carry to decrement operation.
6) Increment contents of register B.
7) Decrement contents of register C.
8) Jump if not zero to rotate the data of A.
9) Move contents of B in A.
10) Store the content at location 2200.
11) Terminate the program.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	~~MVI C~~ LDA 2100	3A	
	2001		00	
	2002		21	
	2003	MVI B, 00	06	
	2004		00	
	2005	MVI C, 08	06	
	2006		08	
	2007	RAL	17	
	2008	JNC 200C	D1	
	2009		0C	
	200A		20	
	200B	INR B	04	
	200C	DCR C	0D	
	200D	JNZ 2007	C2	
	200E		07	
	200F		20	
	2010	MOV A, B	78	
	2011	STA 2200	32	
	2014	HLT	76	

Input/Output: Input: Number at address 2100

Output: The no of one's in the content of 2100 is stored at address 2200

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Experiment No. 17

Objective: Write an assembly language program to find the largest number in an array.

Flow Chart (or) algorithm:

1) Load the address of first number of array in HL pair register.
2) Take counter in register C.
3) Increase the ~~conte~~ register H by 1.
4) Move the content of register H into register B.
5) Decrease the counter.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	MVI C, ~~25~~04	0E	
	2001		04	
	2002	LXI H 3000	21	
	2003		00	
	2004		30	
	2005	MOV A,M	7E	
	2006	INXH	23	
	2007	CMP M	BD	
	2008	JC 200C	D2,0C,20	
	200B	MOV A,M	7E	
	200C	DCR C	0D	
	200D	JNZ 2006	C2	
		~~STA~~	06	
			20	
	200E	STA 3200	32	
	200F		00	
	2010		32	
	2011	HLT	76	

Input/Output: Input: Array of 5 numbers
Output: largest number is stored at location 3200.

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Experiment No. 18

Objective: Write an assembly language program to add 2 8 bit numbers considering carry.

Flow Chart (or) algorithm:

1) Take register C for considering carry.
2) Move first number is register A (accumulator)
3) Move second number in register B.
4) Add the contents of A & B in B.
5) If no carry, jump to step 7.
6) Increase Carry by 1 if carry is generated.
7) Store the result of ~~carry~~ addition at an address is memory.
8) Move the ~~sum~~ content of C to accumulator.
9) Store the result of carry in memory.
10) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	MVI C, 00	0E	
	2001		00	
	2002	MVI A, F3	3E	
	2003		F3	
	2004	MVI B, 15	06	
	2005		15	
	2006	ADD B	80	
	2007	JNC 200B	D2	
	2008		0B	
	2009		20	
	200A	INR C	0C	
	200B	STA 2500	32	
	200C		00	
	200D		25	
	200E	MOV A,C	79	
	200F	STA 2501	32	
	2010		01	
	2011		25	
	2012	HLT	76	

Input/Output: Input :- Numbers - F3 & 15

Output :- Carray is stored in C (01)

Result is stored in address 2500 (08)

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Experiment No. 19

Objective: Write an assembly language program to multiply 2 8 bit numbers.

Flow Chart (or) algorithm:

1) Move data '00' in accumulator.
2) Load Register pair HL from memory location 2700 & 2701 respectively.
3) Add the content of accumulator & register H.
4) Decrement the value of register L.
5) Jump back to Step 3 and 4 if the value of accumulator is non-zero.
6) Store the result at memory location 2500.
7) Stop.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	MVIA ,00	3E	
	2001		00	
	2002	LHLD, 2700	2A	
	2003		00	
	2004		27	
	2005	ADD M	84	
	2006	DCR L	20	
	2007	JNZ 2005	62	
	2008		0005	
	2009		2520	
	200A	STA 2500	32	
	200B		00	
	200C		25	
	200D	HLT	76	

Input/Output: Result of multiplication of 2 numbers is stored at memory location 2500.

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Experiment No. 20

Objective: Write an assembly language program to count even numbers from a series.

Flow Chart (or) algorithm:

1) Store five numbers from location 2000 to 2004.
2) Start the counter using register C as 05.
3) Move the number from memory to accumulator one by one.
4) Perform ANI operation between the content of accumulator and 01.
5) If answer is zero, increase the counter B by 1 everytime we check from series.
6) Store the final result of B at a memory location.
7) Stops the program.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	3000	MVI B, 00	06	
	3001		00	
	3002	MVI C, 05	0E	
	3003		05	
	3004	LXI H 2000	21	
	3005		00	
	3006		20	
	3007	MOV A, M	7E	
	3008	ANI 01	E6	
	3009		01	
	300A	JNZ 300E	C2	
	300B		0E	
	300C		30	
	300D	INR B	04	
	300E	INX H	23	
	300F	DCR C	0D	
	3010	JNZ 3007	C2	
	3011		07	
	3012		30	
	3013	MOV A, B	78	
	3014	STA 3500	32	
	3015		00	
	3016		35	
	3017	HLT	76	

Input/Output: Input: Series of 5 numbers

Output: Total no of even no in a series is stored at location 3500.

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Experiment No. 21

Objective: Write an assembly language to find the average an array in ascending order.

Flow Chart (or) algorithm:

1) Take two counters in B and C for the two loops.
2) Load the content of starting address of array in HL pair register pair.
3) Move the content of HL register pair to accumulator.
4) Increase the value of address of HL pair register pair.
5) Compare the content of HL register pair with memory. (accumulator). [A - M]
6) If number in accumulator is not smaller, swap it with the content of memory.
7) Else, jump to the decrement of counter & decrease counter in C by 1.
8) If the content of accumulator is zero, decrease the counter in B by 1.
9) Else, jump to the point where content of memory is loaded in HL register pair.
10) Repeat the process until all the numbers of an array are compared with all other numbers.
11) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	3000	MVI B ,09	06 09	
	3002	MVI C, 09	0E 09	
	3004	LXI H 2000	7E 00 20	
	3007	MOV A, M	7E	
	3008	INX H	23	
	3009	CMP M	BD	
	300A	JC 3011	DA 11 30	
	300D	MOV D, A	57	
	300E	MOV A, M	7E	
	200F	DCX H	2B	
	3010	MOV M, D	72	
	3011	INX H	23	
	3012 3013	DCR C JNZ 3004	0D C2, 04, 30	
	3016 3017	DCR B JNZ 3007	05 C2, 07, 30	
	301A	HLT	76	

Input/Output: Input : Array of 10 numbers starting from memory location 2000 [10 9 4 2 8 1 3 5 6 7]

Output : Array is arranged in ascending order. [1 2 3 4 5 6 7 8 9 10]

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Experiment No. 22

Objective: Write a program to left shift a number along with carry.

Flow Chart (or) algorithm:

1) Load the content of a memory address is ~~memory~~ accumulat - 04 (05)
2) Shift operation is performed by RAL instruction.
3) The result is stored at memory location 3001 (0A)
4) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	LDA 3000	3A	
	2001		00	
	2002		30	
	2003	RAL	17	
	2004	STA 3001	32	
	2005		01	
	2006		30	
	2007	HLT	76	

Input/Output: Input : 05
Output : 0A

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Experiment No. 23

Objective: Write a program to check whether a number is even or odd.

Flow Chart (or) algorithm:

1) Load the content of memory address (2000) in ~~accumulator (02)~~ HL register pair (02)
2) Move it to accumulator.
3) AND the number with 01.
4) If the output is zero, print / store 22 at a memory address (2500) ie the number is even and stop.
5) Otherwise store 11 at a memory address (2500) ie the number is odd and then jump to ~~stop~~ stop.

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	LXIH 2000	21	
	2002 2003		00 20	
	2004	MOV A, M	4E	
	2005	ANI, 01	EC	
	2006		01	
	2007 2008	JZ, 200F	CA 0F	
	2009		20	
	200A	MVIA, 11	3E	
	200B		~~11~~ 11	
	200C 200D	JMP, 2010	C3 10	
	200E 200F	MVIA, 22	20 3E	
	2010		22	
	2011 2012	STA, 2500	32 00	
	2013		25	
	2014	HLT	76	

Input/Output: Input : 2000 → 02

Output → 22 is stored at memory location 2500.
(11 for odd ; 22 for even)

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Experiment No. 24

Objective: Write a program to check whether a number is negative or positive.

Flow Chart (or) algorithm:

1) Take 0 as value in a register (B).
2) Input a number in another register.
3) Rotate the number without carry.
4) If carry is generated, move the number in B to accumulator and store it in memory (0 ie -ve number)
5) If carry is not generated, increase the value in register B by 1.
6) Move the value in B to accumulator.
7) Store the value of B in memory (1 ie +ve number).
8) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	MVI B, 00	06	
	2001		00	
	2002	MVI A, FD	3E	
	2003		FD	
	2004	RLC	07	
	2005	JC 2009	DA	
	2006		09	
	2007		20	
	2008	INR B	04	
	2009	MOV A, B	78	
	200A	STA 2500	32	
	200B		00	
	200C		25	
	200D	HLT	76	

Input/Output: Input : FD

Output : 0 (Negative)

Note :- 0 for negative & 1 for positive

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Experiment No. 25

Objective: Write a program to count odd numbers from a series.

Flow Chart (or) algorithm:

1) Load the content (data) from memory into HL register pair.
2) Move 0 in register C (counter)
3) Move 0A in register D. (size of series)
4) Increment the content of HL register pair.
5) Move the content of HL register pair to accumulator.
6) AND it with 1.
7) If the result is not zero, decrement the content of register D.
8) If the result is not zero, repeat the process after incrementing C and decrementing the content in D.
9) If the content of D is not zero, move the value in C to accumulator.
10) Store the result in memory.
11) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	LXIH 2500	21,00,25	
	2003	MVIC, 00	0E,00	
	2005	MVID, 0A	16,0A	
	2007	INXH	23	
	2008	MOV A, M	7E	
	2009	ANI, 01	E6,01	
	200B	JZ 200F	CA,0F,20	
	200E	INR C	0C	
	200F	DCR D	15	
	2010	JNZ 2007	C2,07,20	
	2013	MOV A,C	79	
	2014	STA 3050	32,50,30	
	2017	HLT	76	

Input/Output: Input: Content at memory location (07 04 03 08 09 01 02 05 06 00

Output : (stored at location 3050) 5

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Experiment No. 26

Objective: Write an assembly language program to find smallest number from an array.

Flow Chart (or) algorithm:

1) Load the size of array in accumulator from memory.
2) Move the content of accumulator to register C.
3)
4) Load the content of memory in HL register pair (starting address of array)
5) Compare the content of memory with accumulator.
6) If the result doesn't contains carry, jump to statement 8.
7) If no carry is generated, move the content of memory to accumulator.
8) Increment the content of HL register pair by 1.
9) Decrement the content of C by 1.
10) If carry is not generated, jump to statement 5.
11) If carry is generated, store the result at a memory location.
12) Stop

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Program:

Label	Memory Address	Mnemonics	Hex Code	Comment
	2000	LDA 3000	3A, 00, 30	
	2003	MOV C, A	4F	
	2004	XRA A	AF	
	2005	LXI H 3050	21, 50, 30	
	2008	CMP M	BD	
	2009	JNC 200D	D2, 0D, 20	
	200C	MOV A, M	7E	
	200D	INXH	23	
	200E	DCR C	0D	
	200F	JNC 2008	D2, 08, 20	
	2012	STA 2500	32, 00, 25	
	2016	HLT	76	

Input/Output: Input : An array starting from 3050

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